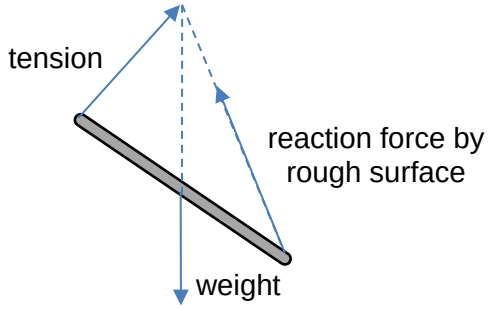
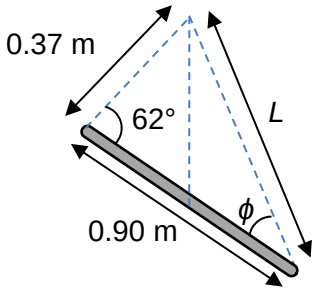
JC2 Preliminary Examinations Paper 3 Mark Scheme

Qn			Answers	Marks
1	(a)		The principle of moments states that if a body is in equilibrium, the sum of all the clockwise moments about any axis/point must be equal to the sum of all the anticlockwise moments about the same axis/point.	B1
	(b)		 To ensure that the lines of action approximately pass through same point.	B1
	(c)		 Recognise that the point of intersection of all 3 forces meet at the fixed point. (rule of concurrency) Let L be the distance from the fixed point to the bottom end of rod Use cosine rule to find distance to top $L^2 = (0.90)^2 + (0.37)^2 - 2(0.90)(0.37)\cos(62^\circ)$ $L = 0.796271$ m	B1
			Use sine rule to find angle ϕ $\frac{\sin\phi}{0.37} = \frac{\sin 62^\circ}{0.796271}$ $\phi = 24.3^\circ$ ∴ angle by ground on rod = 40° + 24.3° = 64.3° (shown)	B1
	(d)		By taking moments about the upper end of rod, (0.200)(9.81)(0.45) cos 40° = F_{ground} (0.90)(sin 24.3°) F_{ground} = 1.82 N	M1 A1
	(e)		All four forces will have their lines of action still meeting at the fixed	B1

			point that the string is attached to.	
			Applying the principle of moments to take moments about that point, the reaction force by the rough surface must continue to point at the fixed point, such that there is no net moment about that point. Thus, angle in **(c)** would remain the same.	B1
	(f)		If the horizontal surface were frictionless, the surface can only exert a force that is vertically upward. Since weight also acts vertically downward, the vector sum of these 2 forces do not have a horizontal component.	M1
			Thus, the force from the string must also have no horizontal component and the string would be vertical. The angle between the string and rod must increase.	A1

Qn			Answers	Marks
2	**(a)**		Newton's Law of Gravitation states that the force of attraction between any two **point** masses is **directly** proportional to the product of their masses and **inversely** proportional to the square of their separation.	B1
	(b)	**(i)**	$F_{Earth} = \frac{GMm}{r^2} = \frac{6.67 \times 10^{-11}(5.98 \times 10^{24})6200}{(1.50 \times 10^{9})^2}$	M1
			$F_{Sun} = \frac{GMm}{r^2} = \frac{6.67 \times 10^{-11}(1.99 \times 10^{30})6200}{(1.50 \times 10^{9} + 1.496 \times 10^{11})^2}$	M0
			Resultant force = gravitational force by earth + gravitational force by sun = 1.10 + 36.0	M1
			= 37.1 N	A1
	(b)	**(ii)**	$F_{net} = mr\omega^2$ $37.1 = 6200(\underbrace{1.50 \times 10^{9} + 1.496 \times 10^{11}}_{1.511 \times 10^{11}})\omega^2$ [Allow ECF from (b)(i)]	M1
			$\omega = 1.99 \times 10^{-7}$ rad s^{-1}	A1
	(b)	**(iii)**	The satellite remains at a **constant** distance of 1.50×10^{9} m from the Earth, on the line joining the centres of the Earth and Sun. (The satellite, Earth and the Sun are collinear).	A1
	(c)		Gravitational force provides the centripetal force.	

			$\frac{GMm}{r^2} = m\frac{v^2}{r}$ $v = \sqrt{\frac{GM}{r}} = \sqrt{GM}\ r^{-\frac{1}{2}}$	M1
			$k = \sqrt{GM}$	A1
			$n = -0.5$	A1
	(d)		When the speed v of the satellite decreases, the centripetal force $\frac{mv^2}{r}$ required for a circular orbit of the same radius r decreases.	B1
			The gravitational force $\frac{GMm}{r^2}$ is initially unchanged and is therefore greater than the centripetal force $\frac{mv^2}{r}$. As a result, there is an unbalanced inward force acting on the satellite in the direction towards the centre of the Earth which cause the satellite to move towards the Earth (distance between the satellite and the Earth decreases).	B1

Qn			Answers	Marks
3	**(a)**	**(i)**	Light from different parts across the slit travels paths of different lengths to a point on the screen, resulting in phase differences. OR Waves will diffract when they exit the slit and spread out as they travel from the slit to the screen. With different path lengths resulting in phase differences.	B1
			When the phase difference of the light is 0, they will interfere constructively and a maximum will be produced. When the phase difference of the light is π rad, they will interfere destructively and a minimum will be produced. OR if path difference, to be in terms of integer of wavelength.	B1
	(a)	**(ii)**	As θ is increased, the angular separation of the two diffraction patterns increases so they change from **unresolved to resolved / distinguishable.**	B1
			The patterns are just resolved when the central maximum of one coincides with the first minimum of the other; beyond this they can be distinguished.	B1
	(a)	**(iii)**	$\theta = \frac{\lambda}{b}$	

			$=\frac{590\times10^{-9}}{0.60\times10^{-3}}$	M1
			$=9.83\times10^{-4}$ rad	A1
	(b)	**(i)**	For 2nd order maximum of red light, $d\sin\theta = n\lambda$ $\frac{1\times10^{-3}}{500}\sin\theta = 2\left(650\times10^{-9}\right)$	M1
			$\theta = 40.5^{\circ}$	A1
		(ii)	For 3rd order maximum of violet light, $\frac{1\times10^{-3}}{500}\sin\theta = 3\left(350\times10^{-9}\right)$ $\theta = 31.7^{\circ}$	A1
			Therefore since the angular position of the 3rd order maximum of violet light is less that of 2nd order maximum of red light, this means that they will overlap.	A1
	(c)		Greater angular separation (greater dispersion), making the wavelengths easier to distinguish / Lower percentage	B1
			Second-order spectrum is less intense (fainter).	B1

Qn			Answers	Marks
4	**(a)**		There is <u>no path for charge to accumulate on the wire connecting the two capacitors</u>. Therefore, any charge transferred onto one capacitor results in an equal amount of charge being transferred onto the other capacitor. Alternative answer: Since the capacitors are connected in series, the <u>same current flows through both capacitors</u> at any instant. The charge transferred is equal to the area under a current–time graph. Hence, the same amount of charge is transferred to both capacitors, so the charge stored on the two capacitors is equal.	B1
	(b)		To determine effective capacitance C_T, $\frac{1}{C_T} = \frac{1}{2.0\times10^{-3}} + \frac{1}{3.0\times10^{-3}}$ $C_T = 1.2\times10^{-3}$ F Magnitude of charge stored for both capacitors must be equal. $C = \frac{Q}{V} \rightarrow Q = CV$ $C_3V_3 = C_TV_T$ $V_3 = \frac{C_T}{C_3}V_T = \left(\frac{1.2\times10^{-3}}{3.0\times10^{-3}}\right)(6.0)$ $= 2.4$ V	M1 M1

	(c)		$Q = CV = (3.0 \times 10^{-3})(2.4)$ $= 7.2 \times 10^{-3}$ C	M1
	(d)		$Q = Q_0(1 - e^{-\frac{t}{RC}}) \rightarrow \frac{Q}{Q_0} = 1 - e^{-\frac{t}{RC}} \rightarrow e^{-\frac{t}{RC}} = 1 - \frac{Q}{Q_0}$ $\rightarrow -\frac{t}{RC} = \ln\left(1 - \frac{Q}{Q_0}\right)$ $t = -RC\ln\left(1 - \frac{Q}{Q_0}\right) = -(400)(1.2 \times 10^{-3})\ln\left(1 - \frac{4.0 \times 10^{-3}}{7.2 \times 10^{-3}}\right)$ $= 0.39$ s	M1 A1
	(e)		$U = \frac{1}{2}\frac{Q^2}{C}$ $U_f - U_i = \frac{1}{2C}(Q_f^2 - Q_i^2)$ $= \frac{1}{2(3.0 \times 10^{-3})}((4.5 \times 10^{-3})^2 - (2.5 \times 10^{-3})^2)$ $= 2.3 \times 10^{-3}$ J	M1 A1
	(f)		The <u>potential difference</u> across the capacitor <u>increases</u> as charges accumulates.	B1
			The <u>energy</u> required to transfer an additional small unit of charge is <u>proportional</u> to the <u>potential difference</u>, so more energy is required to transfer each successive unit of charge. ($\Delta W = V\,\Delta Q$)	B1
			Therefore, <u>more work is done per unit charge</u> to move further charge into the capacitor.	A0

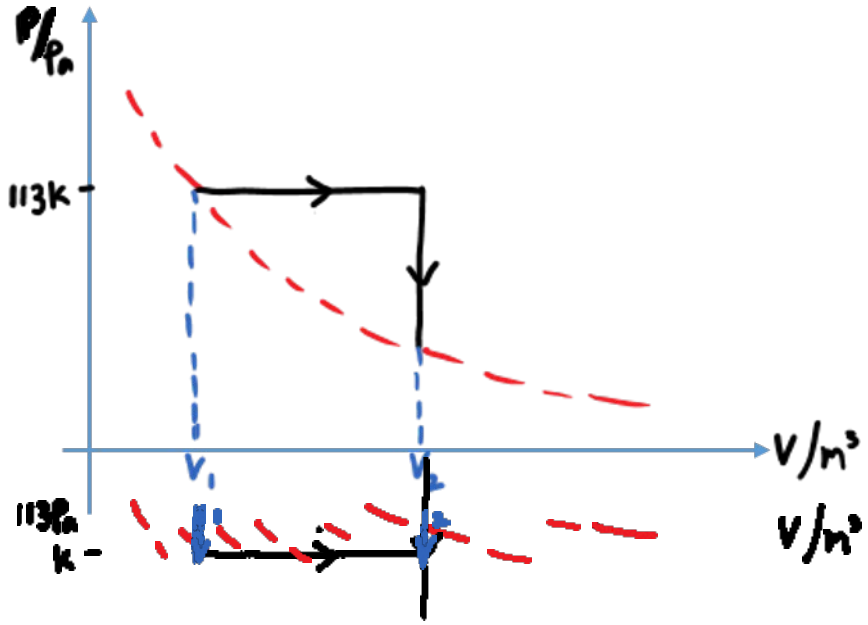
Qn			Answers	Marks
5	**(a)**		Each <u>transition</u> between electron energy levels involves the <u>emission or absorption of one photon</u>. The energy of the photon corresponds to the energy difference between the two levels.	B1
			Line spectra consist of discrete lines, showing that atoms only <u>emit or absorb photons with specific energies / frequencies</u>. This provides evidence that electrons in isolated atoms occupy discrete energy levels.	B1
	(b)	**(i)**	Difference in energy of first excited state and the ground state corresponds to the photon absorbed with the lowest energy (longest wavelength). energy of photon $= \frac{hc}{\lambda}$ energy of photon $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{500 \times 10^{-9}} = 3.978 \times 10^{-19}$ J	M1
			Hence, energy of first excited state is…	A1

			3.978×10^{-19} J $= E_2 - E_1$ $E_2 = -7.60 \times 10^{-19} + 3.978 \times 10^{-19}$ $= -3.62 \times 10^{-19}$ J	
	(b)	**(ii)**	The reason why the emission spectrum has one more line is due to the transition from E_3 to E_2. $E_3 - E_2 = \frac{hc}{\lambda_{1\to 3}} - \frac{hc}{\lambda_{1\to 2}} = hc\left(\frac{1}{300\times10^{-9}} - \frac{1}{500\times10^{-9}}\right)$ energy of photon $= \frac{hc}{\lambda_{3\to 2}}$ $\cancel{hc}\left(\frac{1}{300\times10^{-9}} - \frac{1}{500\times10^{-9}}\right) = \frac{\cancel{hc}}{\lambda_{3\to 2}}$	M1
			$\lambda_{3\to 2} = 7.50 \times 10^{-7}$ m	A1

Qn			Answers	Marks
6	**(a)**		It is the probability per unit time of a decay.	B1
	(b)	**(i)**	half-time = 14 mins = 840 s	B1
			activity = $\frac{\ln 2}{T_{1/2}} N = \frac{\ln 2}{840} \times 2.0 \times 10^7$	B1
			= 16500 Bq or 17000 Bq (2 sf)	A1
			alternative	
			activity = gradient of N - t graph	B1
			method to find gradient - Correct coordinates - Correct formula	M1
			A between $\pm 10\%$ of 16500 Bq (14900 Bq to 18200 Bq)	A1
		(ii)	half-life is short, so no time for technetium-101 to be delivered into tumour or half-life in short implies activity is high during delivery to tumour. So another organs may be damaged during delivery / too low at the end and therefore ineffective or half-life is short, so no time to be technetium-101 to be delivered to hospital after production or	B1

			half-life is short, power changes quickly over time. Cannot estimate the dose.	

Qn			Answers	Marks
7	(a)	(i)	The **specific heat capacity**, *c*, of a body is defined as the amount of thermal energy required to raise the temperature of unit mass of the substance by one degree. The S.I. unit is J kg^{-1} K^{-1}.	B1
	(a)	(ii)	The same steady inflow and outflow temperatures are maintained for both experiments so that the rate of heat lost is the same in both experiments so that it can be taken into account in the conservation of energy equations when calculating the specific heat capacity.	B1
	(a)	(iii)	$V_1 I_1 = \frac{m_1}{t_1} c\Delta\theta + \frac{H_1}{t_1}$ ---------- (1) $V_2 I_2 = \frac{m_2}{t_2} c\Delta\theta + \frac{H_2}{t_2}$ ---------- (2) To award one mark if students managed to relate power to rate of energy absorption, but missed out on heat loss.	M1
			Since $\frac{H_1}{t_1} = \frac{H_2}{t_2}$, $c = \frac{(V_2 I_2 - V_1 I_1)}{(\frac{m_2}{t_2} - \frac{m_1}{t_1})\Delta\theta} = \frac{(37.8 - 25.2)(60)}{(0.232 - 0.150)(17.4 - 15.2)}$	M1
			= 4190 J kg^{-1} K^{-1}	A1
	(b)	(i)	1. Isobaric expansion / expansion under constant pressure	B1
			2. Isovolumetric cooling / cooling under constant volume	B1
	(b)	(ii)	F = mg = 8.00 (9.81) = 78.5 N	M1
			P = 78.5/(60.0 x $10^{-4)}$ + 100000 Pa	C1
			= 1.13 x 10^5 Pa	A1

	(b)	**(iii)**	 (Not necessary to draw in the isotherm.)	
			-1 mark for expansion under constant pressure (horizontal to the right)	B1
			-1 mark for cooling under constant volume (vertical line downwards)	B1
			Lose second mark if line is disjointed, or not in correct order.	
	(b)	**(iv)**	For heating, $\Delta U = Q_1 + W = Q_1 + p\Delta V$ For cooling, $W = 0$ $\therefore \Delta U = Q_2$	B1
			Since ΔU is the same for both processes, $Q_1 + p\Delta V = Q_2$ Hence net heat gained by gas = $Q_1 - Q_2$ $= p\Delta V$ $= 113000 \times 0.200 \times 60.0 \times 10^{-4}$ $= 136$ J OR $\Delta U_{net} = Q_{net} + W_{net}$ Since there is no change in the temperature between (a) and (c), $\Delta U_{net} = 0$ $Q_{net} = - W_{net}$ $= 1.13 \times 10^5 \times 0.200 \times 60.0 \times 10^{-4}$ $= 136$ J	M1 M1 A1 B1 M1 M1 A1
	(c)	**(i)**	As the air expands, its <u>pressure decreases</u>. The lower air pressure produces a <u>smaller downward force</u> on the water.	B1
			According to Newton's third law of motion, the water exerts an <u>equal and opposite smaller force on the bottle</u>, causing it to move	B1

			upwards.	
	(c)	**(ii)**	Select two suitable points from the graph, for example: $p_1 \approx 3.6\times10^{5}$ Pa, $V_1 \approx 1.1\times10^{-3}$ m^3 $p_2 \approx 1.0\times10^{5}$ Pa, $V_2 \approx 3.0\times10^{-3}$ m^3 $p_1V_1 \approx 4.0\times10^{2}$ J $p_2V_2 \approx 3.0\times10^{2}$ J.	M1
			Therefore, pV **decreases** during the expansion. Since the amount of gas is constant and $pV = nRT$, the temperature of the air **decreases** as the air expands.	A1

Qn			Answers	Marks
8	**(a)**	**(i)**	Electric field is directed downwards; for the oil drop to remain stationary / Electric field points in direction of decreasing potential	M1
			the electric force must act upwards, hence charge is negative	A1
		(ii)	$E = \frac{V}{d} = \frac{680}{7.0\times10^{-3}} = 9.71\times10^{4}$ V m^{-1}	C1
			$qE = 4.8\times10^{-14}$ $q = \frac{4.8\times10^{-14}}{9.71\times10^{4}}$	M1
			$q = -4.94\times10^{-19}$ C e.c.f from (a)(i) for sign of charge	A1
		(iii)	$m = \frac{4.8\times10^{-14}}{9.81} = 4.89\times10^{-15}$ kg $F_E - W = ma$ $\frac{qV}{d} = W + ma$	M1
			$V = \frac{(W+ma)d}{q}$ $V = \frac{(4.8\times10^{-14} + 4.89\times10^{-15}\times0.50)(7.0\times10^{-3})}{4.94\times10^{-19}}$	M1
			$= 710$ V $\quad (714.8)$	A1
	(b)	**(i)**	If *B* is into the page, then *E* is downwards. OR If *B* is out of the page, then *E* is upwards	A1

		(ii)	$E = vB = \left(1.6 \times 10^{5}\right)\left(9.7 \times 10^{-2}\right)$	M1
			$E = 1.55 \times 10^{4}\ \text{V m}^{-1}$	A1
		(iii)	Particles will experience a <u>larger electric force</u>, only particles that <u>move faster will have a larger magnetic force</u> such that net force is zero.	M1
			Therefore, particles with <u>a greater speed pass undeviated</u>.	A1
	(c)	**(i)**	A <u>current-carrying wire in a magnetic field</u> experiences <u>a magnetic force. / Interaction between the magnetic field produced by the current and that of the magnet results in a force between the wire and magnet.</u>	B1
			The magnet experiences a <u>upward force by the wire,</u> <u>reducing the force exerted on the balance</u> and hence the balance reading.	B1
		(ii)	**1.** For the magnet in equilibrium, $W = F_B + N$ $W = BL\left(2.0\right) + 2.20$ $W = BL\left(3.5\right) + 2.14$	C1
			$BL = 0.04\ \ N\ \ A^{-1}; W = 2.28\ \ N$	C1
			$M = 0.232\ \ kg$	A1
			2. The <u>magnetic force on wire is downwards</u> according to Newton's 3rd law. By <u>Fleming's left hand rule</u>, the direction of the current	M1
			must be from B to A.	A1